

RESEARCH ARTICLE

SmartLaw Advisor: AI-Powered Legal Consultation and Contract Analysis for Saudi Arabia

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ABSTRACT The complexity of local laws and the scarcity of affordable legal support in Saudi Arabia make discerning legal issues a challenge. The scarcity of Arabic digital assistants and the absence of innovative platforms compatible with Saudi legislation compound this challenge. We present an innovative virtual legal advisory platform to enhance access to legal help using state-of-the-art language models. The system is designed around two main functions: 1) a legal chatbot built on Google's Gemini 1.5, with a standard RAG (Retrieval-Augmented Generation) pipeline and a custom Arabic legal corpus based on official Saudi regulations; and 2) a contract analysis system that extracts and classifies Arabic legal documents to determine their compliance with national law. With a simple RAG formulation, the model retrieves relevant legal texts via semantic matching and produces precise, regulation-aligned answers without needing to retrain it. The system was evaluated on 150 legal questions across ten domains and tested on 50 real contract drafts. Gemini outperformed models such as GPT-4, GPT-3.5, Command-A, and ALLaM in terms of accuracy (98.67%), response speed, and expert evaluation. This work demonstrates how AI can provide affordable and efficient legal advice in Arabic and contribute to Saudi Arabia's digital transformation and Vision 2030. The system was further validated using a legally verified Saudi corpus and an expert-reviewed dataset comprising 150 legal questions and 50 contract samples, ensuring methodological rigor and dataset reliability.

INDEX TERMS AI law, artificial intelligence, legislation, contract analysis, legal advice.

I. INTRODUCTION

Greatly influenced different fields of life, such as education, medicine, and commerce. Its advanced data analysis and optimization capabilities are heavily used to increase business efficiency and deliver better results. This is strategic for the Kingdom of Saudi Arabia. On the other hand, AI is anticipated, in accordance with the national Vision 2030 and the plan to evolve the national economy into a knowledge-based one, as a key facilitator. AI is the key to driving about 70% of the objectives of the Vision, according to the Saudi Data and Artificial Intelligence Authority (SDAIA) [1].

Given these critical developments and the Kingdom's clear prioritization of AI, the use of AI in the Saudi Arabian legal market is still relatively nascent. Technology has yet

to make inroads into legal procedures and consultancies. As a result, despite its enormous potential to transform the field, there is a striking disparity due to a lack of accessible, robust, user-friendly applications created for research in this area. Furthermore, the Saudi legal services market is valued at over SAR 8.5 billion annually, reflecting an increasing public and commercial need for accessible legal guidance. While several AI-driven legal platforms exist globally such as DoNotPay, LegalMation, Harvey AI, and ChatLaw—none provide Arabic-language support or alignment with Saudi statutory and regulatory frameworks. However, their architectures and datasets are tailored to non-Arabic jurisdictions and cannot interpret Arabic legal morphology or Saudi-specific regulatory frameworks. Unlike these systems, our work introduces a Saudi-focused RAG pipeline, an Arabic semantic chunking workflow, and a dual-architecture system integrating both legal consultation and contract analysis,

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collectively representing a novel contribution not found in existing legal chatbots. This gap highlights the need for localized AI systems capable of understanding Arabic legal text and the unique structure of Saudi law. Despite progress in global legal AI, significant gaps remain in the Arabic and Saudi legal domains. There is no publicly available Saudi-focused RAG dataset, and Arabic legal retrieval benchmarks are minimal due to restricted access to machine-readable legal corpora. Existing LLM-based legal assistants do not support Arabic semantic chunking, lack alignment with Saudi statutory structures, and provide no integrated framework that combines contract analysis with legal consultation. Additionally, prior work does not evaluate LLM performance on Arabic legal reasoning using expert-reviewed datasets. These gaps highlight the need for a localized, rigorously evaluated Arabic legal AI system such as the one proposed in this work.

The high cost of conventional legal services is a significant obstacle, as it becomes increasingly inaccessible to the majority of people, especially those with low incomes, even in cases that require advice that could be provided in just a few minutes. This economic barrier frequently leaves members of the public without critical legal information they need to resolve disputes or to understand their rights, which can result in negative consequences, including unwise decisions or the inability to exercise their rights. The problem is further exacerbated in contractual contexts, where the weaker of the two parties is misled or deceived into signing an unfavorable contract. The high cost of professional legal review discourages both consumers and businesses from obtaining this critical service and therefore increases the likelihood that fraud will occur.

Although promising experiments worldwide, such as AI-based dispute resolution in Finland and contract analysis models in Italy, have demonstrated the feasibility of these solutions, they often suffer from being tied to a specific legal system, hindering agility and adaptability to local legal specifics. Consequently, a significant gap exists in the Saudi context. Within Saudi Arabia, there is no advanced platform that can provide precise, low-cost legal advice and contract analysis in Arabic while fully satisfying national law requirements.

This absence represents a serious obstacle to justice, particularly for average citizens and small business proprietors who cannot always afford traditional legal assistance. When contracts are not reviewed, there is no opportunity to prevent legal disputes or fraud. To bridge this gap, this research aims to leverage AI technologies to provide accurate and accessible legal consultations and contract analysis tailored to the Saudi legal system. This paper directly addresses this gap by developing a smart virtual legal advisory platform. The primary objectives of this research are to study the effectiveness of AI in providing reliable legal consultations, offer a low-cost service, reduce contract fraud through automated reviews, and enhance legal awareness.

A. RESEARCH CONTRIBUTION

To achieve the objectives, the research introduces a system with the following main contributions, and supported by:

- 1) A legal chatbot built on a Retrieval-Augmented Generation (RAG) pipeline with a custom Saudi legal corpus.
- 2) A contract analysis tool for Arabic documents. The contributions of this work include the implementation and rigorous evaluation of the integrated platform, demonstrating its high accuracy and potential to provide accessible legal support aligned with Saudi Arabia's digital transformation goals.
- 3) We developed the first structured Arabic-language Saudi legal corpus and an associated evaluation dataset, covering statutory texts, regulatory provisions, and real contracts.
- 4) We propose an integrated architecture combining a RAG-powered legal consultation system with an Arabic-optimized contract analysis module, tailored to the linguistic and jurisprudential characteristics of Saudi legal texts.
- 5) We conduct comparative experiments involving Gemini 1.5 Flash, GPT-4, GPT-3.5, ALLaM, and Command-A, offering empirical insights into LLM performance on Arabic legal reasoning—an underexplored research area.
- 6) Saudi legal specialists evaluated the system outputs for correctness, clarity, and alignment with national laws, providing practical assessment beyond automated metrics.

II. PROBLEM STATEMENT

Saudi Arabia faces a wide gap between the innovation potential of artificial intelligence (AI) and its actual application in the field of law [2], which poses a significant obstacle to justice in the country that obstructs the Vision 2030 national plan to transform the economy into a digital, knowledge-based society. Although the strategic framework for national transformation incorporates the aim of AI [1], it has not yet been practically applied in legal processes in a meaningful way [3]. This discrepancy appears in two crucial and interrelated issues: accessibility and localization. The high price of traditional legal services precludes most people and companies from obtaining critical counsel, leading to unresolved disputes, rights forfeited, and uninformed decisions, while rendering them more vulnerable to contract fraud because the expense of professional review discourages all but the most confident from vetting their agreements. However, while encouraging examples such as applications in Finland and Italy have shown the potential of AI in legal domains such as dispute resolution and contract analysis, these are limited by being tied to specific foreign legal systems and lacking the required nuance of the Arabic language. As a result, there is a significant gap in the provision of such services in Arabic for the Saudi legal system, which is based in part on Sharia (Islamic law) and

statutory law. There is no intelligent platform available that can deliver cost-effective, accurate legal advice and in-depth contract analysis for this legal system [4]. This absence thus not only reinforces inequality of access to justice but also underlines the pressing need for a technically feasible, accessible, and fine-tuned solution.

III. RESEARCH QUESTIONS

After considering the above discussion and our understanding of the problem space, the following research questions are posed:

RQ1: Can artificial intelligence provide reliable legal consultations? **RQ2:** How accurate is artificial intelligence in analyzing contracts and identifying unsuitable terms? **RQ3:** Are users willing to rely on an AI-based legal platform?

In addition to the above questions, this study also investigates the challenges that such a platform might encounter in the legal domain.

IV. AIM AND OBJECTIVES

The aim of this research is to leverage AI technologies to deliver accurate and accessible legal consultations and to perform contract analysis in alignment with the Saudi legal system. To realize this aim, the following objectives are defined:

- To investigate the effectiveness of AI in offering reliable legal consultations (RQ1).
- To provide a low-cost legal service accessible to all (RQ1).
- To reduce occurrences of contract fraud via automated contract reviews (RQ2).
- To enhance legal awareness by guiding users regarding their legal rights and obligations (RQ3).

Driven by these objectives, this work proposes an intelligent legal platform powered by advanced AI models. Specifically, it incorporates a Retrieval-Augmented Generation (RAG) pipeline paired with a custom Saudi legal corpus to support context-aware legal advice, alongside a contract analysis tool for Arabic documents that extracts, evaluates, and flags compliance issues.

The rest of the paper is structured as follows: Section V presents related work focusing on Saudi laws and Arabic-language systems. Section VI details the methodology and explains semantic retrieval through a multi-stage pipeline. Section VII outlines the system architecture with insights into both the legal advice and contract analysis modules, including algorithmic complexity. Section VIII describes the experiments, followed by a Results section presenting outcomes, and the Conclusion and Future Work sections at the end.

V. LITERATURE REVIEW

Initial work in legal artificial intelligence (AI) mainly explored low-cost legal platforms for general legal tasks [5]. In that context, the ChatLaw project (2023) [6] introduced

a legal language model built upon an open-source engine, assisting legal professionals in advanced text processing. This effort established an early benchmark for integrating AI into law and demonstrated the usefulness of specialized legal corpora in language models [7].

Subsequently, research shifted toward developing intelligent agents capable of understanding and processing users' questions in real time. One notable example is the AI-based Legal Virtual Assistant (2024) [8], which uses a RAG-aided large language model to assist in resolving housing disputes in Finland. This marks progression toward interactive, dialogue-driven advisory systems—albeit within restricted jurisdictions and domains. Concurrently, initiatives emerged to explore how AI can enhance legal reasoning. The AI Assistance in Legal Analysis experiment [9] studied AI's role in supporting human legal interpretation, showing promising results for scholarly and litigation uses. Later, DISC-LawLLM (2023) [10] applied domain-specific fine-tuning techniques to boost performance on legal datasets. Other contributions addressed reliability and ethics; for instance, the Hallucination-Free study (2024) [11] examined factual inaccuracies in AI-generated legal texts, while the Artificial Intelligence and the Legal Profession project [12] discussed issues of transparency, accountability, and trust in automated legal systems. Collectively, these efforts underscore the tension between deploying AI capabilities and preserving legal integrity [13].

Parallel to these developments, significant interest has focused on AI for contract analysis and management [14]. Benchmark studies such as “Comparing the Performance of AI to Human Lawyers in Reviewing Standard Business Contracts” [15] empirically compared AI against lawyers on metrics like efficiency, accuracy, and cost—finding that AI can in some tasks surpass human-level performance. Later works, including Extraction from Contracts Using LLMs [16], demonstrated how LLMs extract structured data from contracts, streamlining legal workflows. Additional studies (e.g. [17]) explored machine learning and NLP approaches to analyze legal documents and support decision-making in contract contexts.

Meanwhile, efforts to enhance document accessibility have gained attention. Automated Summary of Legal Texts [18] proposes methods for condensing complex legal texts while preserving neutrality and jurisprudential meaning. Initiatives like Towards Developing AI-Based Semantic Models for Historical Legal Documents [19] aim to improve interpretability of legal language in legacy texts. In the Saudi context, the Tawqe platform [20] is among the earliest local attempts to deploy AI for contract generation and analysis. Though a commendable step, it is limited to automating document processing rather than providing full advisory functionality. Recent Gulf-region initiatives have explored AI applications in legal and governmental contexts, including Saudi Arabia's digital justice transformation programs and the UAE's AI governance frameworks. Studies such as [21], [22] on AI

readiness in Saudi legal services and [23] on AI adoption in Gulf e-government systems highlight the regional momentum toward intelligent legal automation. However, these works do not provide Arabic-language legal RAG systems or contract-analysis pipelines. This gap further motivates the need for a localized, Saudi-specific legal AI framework.

Despite these advances, several major limitations remain [24]. Most systems are narrow in scope—either restricted to particular jurisdictions or designed primarily for document generation rather than deeper analysis. Critically, no platform effectively handles Arabic legal texts, especially within the Saudi legal system where Sharia and statutory law intersect [25]. This gap underscores the urgency for innovation. Some prominent work are further summarised in the Table 1

As will be detailed later in the methodology section, our proposed solution bridges this gap by integrating legal advice and contract analysis into a unified system. The solution is designed to include Chroma, Gemini 1.5 Flash with RAG, and employs a legally validated Saudi corpus to deliver high-accuracy, context-sensitive assistance tailored to Saudi legal norms in Arabic—an approach not previously addressed in the literature.

VI. METHODOLOGY

In this work, we utilize Gemini 1.5 Flash [26], a novel language model published by Google in 2024, which is well-known for its implied real-time and versatility features, as well as improved accuracy in Arabic understanding. Different from standard LLMs, Gemini 1.5 Flash is configured for a low scatter ratio and for processing multimodal inputs—such as text and document-based information—making it well-suited for law-based use-cases covering consultative and structured document analysis. The model was implemented in our system through the Google Generative AI API and enhanced via a Retriever-Augmented Generation (RAG) framework. RAG is a model of a hybrid type that combines information retrieval with generative language modeling. When a user makes a query, the system carries out a matching operation in a Saudi legal statutes and regulations custom corpus. The resulting legal provisions subsequently inform the LLM to produce responses that are linguistically coherent (to date, no questions have been answered) and legally justified, maintaining regulation compliance and context preservation. To clarify the overall architecture and workflow, Figure 1 provides a high-level diagram illustrating the integration of the two main system modules: the Legal QA System (with RAG) and the Contract Analysis System, within the unified platform. Figure 2 provides further details about the model. The methodology incorporates key NLP and RAG components, including semantic chunking, vector embedding generation, dense retrieval, cross-encoder reranking, OCR-driven text normalization, and JSON-structured clause extraction to ensure accurate and legally aligned outputs.

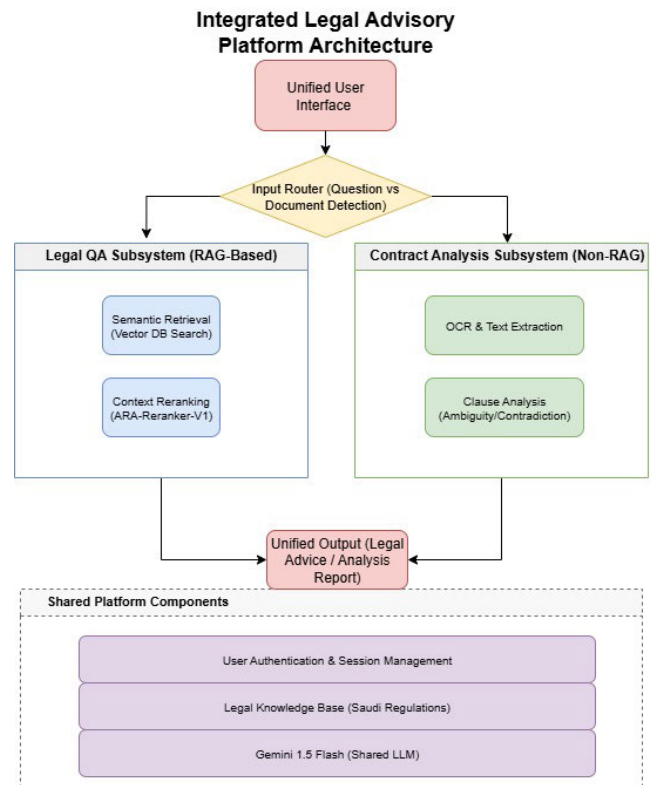


FIGURE 1. Integration of the two subsystems overview.

A. SEMANTIC RETRIEVAL VIA MULTI-STAGE PROCESSING PIPELINE

For accurate semantic retrieval, a multi-stage processing pipeline was implemented. A detailed workflow of this RAG pipeline is depicted in Figure 3. Raw legal documents were first segmented via a semantic chunking algorithm to generate coherent textual units best suited to context-based alignment. These segments were vectorized using GATE-AraBERT-v1, a transformer-based embedding model finetuned on Arabic legal terms, also used in a recent work by [27]. The embeddings were indexed in Chroma, a high-performance vector database fronting for efficient similarity search. At testing, user queries are vectorised by the same embedding model, and the most semantically similar legal segments are retrieved from the database. Later, these candidates are refined using ARA-Reranker-V1 [28]. Keeping their priority in terms of jurisprudential relevance, they undergo contextual compression to extract essential information and remove redundancy and irrelevance before feeding it to the generative model. Crucially, the design of this vector database allows for dynamic updates. The corpus can be refreshed by ingesting new or amended Saudi laws and regulations. The process involves re-running the document ingestion, chunking, and embedding steps for the updated documents, followed by updating the Chroma vector store. This ensures the system's knowledge base remains current with the evolving legal landscape without requiring a full retraining of the language

TABLE 1. Comparison of Existing Legal AI Systems and the Proposed Work.

System	Language	Dataset Source	Tasks Supported	Metrics Used	Limitations vs. Our System
DoNotPay	English	Proprietary case & policy templates	Legal advice generation	Not publicly reported	Not Arabic; no statutory alignment; no contract analysis
LegalMation	English	US civil litigation cases	Document automation; filing generation	Precision/Recall (internal)	Not Arabic; domain-specific only
Harvey AI	English	Large multilingual corporate legal data	General-purpose legal drafting & Q&A	Not publicly disclosed	No Saudi law support; not jurisdiction-specific
ChatLaw	Chinese	Chinese court judgments & legal statutes	Legal Q&A; statute lookup	Accuracy; BLEU	Chinese only; no contract analysis
LawGPT	English	U.S. & EU legal texts	Legal Q&A; summarization	Accuracy (internal)	Not Arabic; cannot process Arabic morphology
Proposed System	Arabic (Saudi)	Saudi statutes, regulations, contracts	RAG-based consultation + contract analysis	Accuracy, expert scoring, legal alignment	First localized system for Saudi law; Arabic semantic chunking; dual architecture

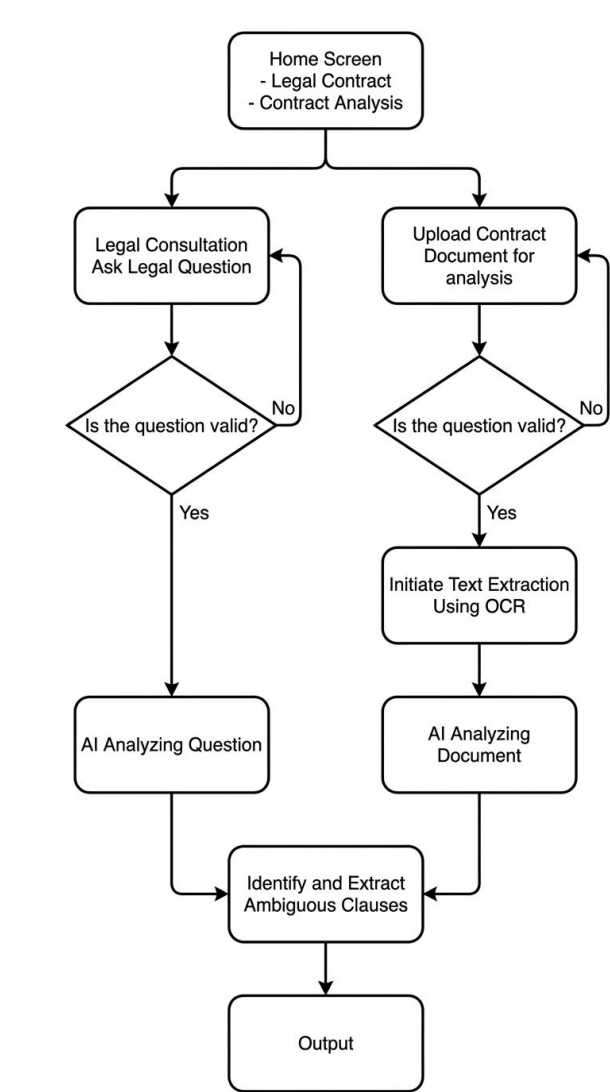


FIGURE 2. High Level Diagram of the Legal QA system and Contract Analysis.

model. In the domain of the legal consulting module, this pipeline enables the provision of contextually accurate and

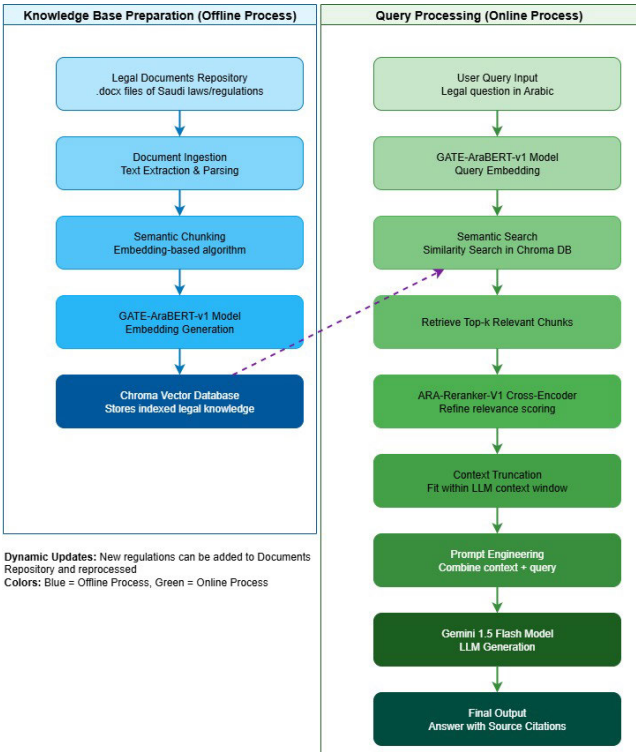


FIGURE 3. Multi Stage Processing Pipeline.

jurisdictionally relevant answers according to Saudi legal norms. For example, when asking the process of ending a contract under Saudi labor law, the user would receive an answer constructed from applicable articles from the official labor code, expressed in lay, yet legally accurate, Arabic.

Within the contract analysis module, users can upload legal documents in PDF format, which are processed through an OCR pipeline to extract machine-readable text for subsequent analysis. The Gemini model then does clause-level analysis to detect non-compliant clauses, regulatory gaps, or ambiguity in expression, and will then produce natural language recom-

mentations in Arabic to alert if any risks can be identified or to recommend changes. The OCR component applies layout-aware text extraction optimized for Arabic legal documents, enabling accurate parsing of structured clauses, headings, and contract-specific terminology. In the context of legal documents, the contract analysis module incorporates OCR technology to process uploaded legal documents while preserving formatting and structural integrity. When a user submits a contract, the system first employs Google's Gemini API multimodal capabilities to extract textual content while preserving critical formatting elements, multilingual content (Arabic/English), and special symbols. The OCR engine is optimized explicitly for legal documents, handling complex layouts, tabular data, and mixed script environments characteristic of Saudi legal contracts. Special attention is given to the accurate extraction of legally significant elements such as dates in DD-MM-YYYY format, monetary figures, and clause numbering systems. For enhanced accuracy with handwritten annotations or poor-quality scans, the system implements preprocessing techniques including image deskewing, noise reduction, and contrast enhancement. The extracted text undergoes validation routines to flag potential OCR errors, particularly for critical legal terms and numerical values, ensuring the integrity of subsequent analysis phases. This robust OCR foundation enables the Gemini model to perform precise clause-level analysis, ambiguity detection, and compliance checking against Saudi legal standards.

This unified dual-pipeline architecture consists of retrieval-based validation and generative response synthesis modules, which allows our system to provide real-time legal consults and detailed contract analysis in an exact manner, at an impressive efficiency and high scalability. The architecture allows for future extension to meet changing regulations and potential extension to other legal areas.

The methodology integrated recent developments in transformer-based LLMs, semantic retrieval, and RAG protocols to develop a digital legal aid platform adapted to the Saudi legal domain. We have chosen Gemini 1.5 Flash due to its Arabic legal language superiority, multimodal flexibility, and strong integration capabilities, rendering it an ideal environment to build regulation-aware, intelligent legal services. The following section provides a detailed description of the model.

B. ETHICAL CONSIDERATION

Ensuring the ethical and responsible deployment of AI is essential when handling sensitive legal information. The proposed system incorporates several safeguards:

- 1) **Data Privacy and Confidentiality:** Users' uploaded contracts are processed in memory and are not stored permanently. All documents are automatically anonymized by removing personal identifiers (names, ID numbers, addresses) before analysis. No user data is

retained outside the session, ensuring compliance with Saudi data-protection regulations.

- 2) **Bias and Fairness Mitigation:** To mitigate potential historical or linguistic bias in Arabic legal corpora, retrieval results were reviewed by legal experts during system development. The RAG pipeline reduces the risk of hallucination by grounding answers strictly in retrieved legal provisions. Model temperature was set to 0.2 to minimize speculative or biased interpretations.
- 3) **Security of Legal Documents:** OCR-processed contracts remain encrypted in transit during PDF upload using platform-provided HTTPS security. No documents are transmitted to third parties or reused for model training.
- 4) **Human Oversight:** The system is not intended to replace lawyers; instead, it assists users by clarifying applicable regulations. All outputs advise users to consult licensed legal professionals for final decisions.

These safeguards ensure that the platform provides transparent, privacy-conscious, and legally responsible AI support.

VII. SYSTEM ARCHITECTURE

This section is a technical overview of two distinct AI systems that are used to analyze legal documents and respond to questions in a specific domain. The former one is to use a separate Large Language Model (LLM) to explore the contracts, and the latter one is an enhancement of the Retrieval-Augmented Generation (RAG) structure to produce a more contextual and source-sensitive legal consultation chatbot. The core objective is to detail the logical design, data flow, and key components of each system without delving into specific code implementations. Document-generation systems (such as contract drafting models) focus on producing new legal text based on patterns learned during training. In contrast, a consultation-level RAG system retrieves relevant statutory or regulatory provisions and grounds the answer directly in authoritative legal sources. This ensures factual accuracy, reduces the risk of hallucinations, and enables precise alignment with Saudi legal requirements. The proposed system belongs to the second category, integrating retrieval with generation to support legally verified consultation. The overall flow of the Architecture user interaction is provided in the Figure 4.

A. SYSTEM 1: PIPELINE FOR ANALYZING CONTRACTS (WITHOUT RAG)

The first system is intended as a batch processing pipeline for contract document analysis, particularly for detecting problematic or ambiguous clauses. It works with no retrieval component, only leveraging the parametric knowledge and reasoning of a pre-trained LLM (Google's Gemini).

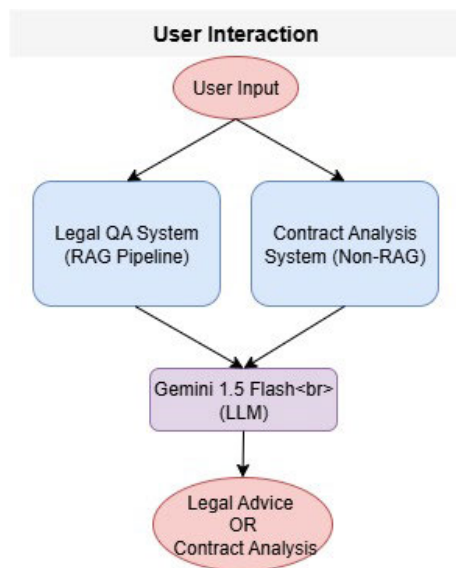


FIGURE 4. Architecture of the Analyzing Contracts and Legal Advice User Interaction.

1) LOGICAL ARCHITECTURE AND WORKFLOW

The system follows a sequential, modular pipeline:

1) PDF Text Extraction:

- **Input:** A file path to a PDF contract.
- **Process:** The PDF is uploaded to the Gemini API via its multimodal capabilities. A precise prompt instructs the model to extract all text faithfully, preserving formatting, multilingual content (Arabic/English), and symbols. Special emphasis is placed on the accurate extraction of dates in a specified format (DD-MM-YYYY).
- **Output:** Raw, extracted text from the contract.

2) Text Preprocessing (Date Validation):

- A validation subroutine scans the extracted text for date strings matching the expected format. Invalid or malformed dates are flagged for logging purposes, ensuring data quality before deeper analysis.

3) Clause Extraction:

- **Input:** The preprocessed contract text.
- **Process:** The Gemini model is prompted again to act as a clause identifier. The prompt provides strict instructions to isolate and list all contractual clauses verbatim, avoiding summarization or paraphrasing. This step effectively segments the monolithic contract text into discrete, analyzable units.
- **Output:** A structured list of extracted clauses.

4) Clause Analysis:

- **Input:** The list of extracted clauses.
- **Process:** This is the core analytical step. The system can be configured for two types of analysis:

- **Ambiguity Analysis:** Gemini is prompted to analyze each clause for vague language, undefined terms, or missing specifics. The prompt mandates a structured JSON output containing the original clause, a description of the identified problem, and a concrete recommendation for clarification.
- **Contradiction Analysis:** Gemini is prompted to compare clauses across the document to identify logical conflicts (e.g., differing delivery timelines, conflicting payment terms). The output is a structured JSON listing the contradictory clause pairs, the nature of the contradiction, and a resolution recommendation.
- **Configuration:** The model's temperature is set very low (0.2) and the response format is constrained to application/json to maximize determinism and output consistency.
- **Output:** A structured JSON object containing the analysis results.

5) Result Aggregation:

- The final results from the analysis step are compiled and returned as the system's output, ready for review by a legal professional.

All extracted text—whether from PDFs, OCR, or statutory documents—undergoes a standardized preprocessing pipeline. Arabic script normalization includes unifying different forms, removing kashida, eliminating diacritics, and normalizing hamza variations. Tokenization is performed using the Gemini 1.5 Flash with an Arabic focus tokenizer with a maximum token window of 1,024 tokens per chunk. Stopwords from Arabic legal vocabulary lists are retained (not removed) to preserve statutory semantics. For contract analysis, OCR outputs are cleaned using rule-based line merging, punctuation restoration, and clause boundary detection before tokenization. This preprocessing ensures consistent embedding generation and legal-context preservation. To be more specific, because many user-uploaded contracts are scanned images, the OCR output may contain character-level and word-level errors. To mitigate this, the system applies an OCR correction layer that includes Arabic character normalisation (e.g., normalising hamza forms, removing tatweel, fixing common OCR confusions), dictionary-assisted correction using a curated list of Arabic legal terms, and rule-based reconstruction of broken words caused by line breaks. Additionally, low-confidence OCR segments are flagged and passed through a second-pass LLM refinement step to restore likely legal terminology. This hybrid correction mechanism significantly reduces OCR noise before clause extraction and analysis. Contract clauses are segmented using a hybrid approach that combines punctuation-based splitting, Arabic legal connective patterns, and rule-based heuristics to detect clause boundaries even when punctuation is inconsistent. Each extracted clause is then assigned to one of the predefined legal categories—such

as Payment Terms, Termination, Obligations, Liabilities, Intellectual Property, or Confidentiality. Categorization is performed using an LLM-guided classifier that evaluates both the semantic content and structural indicators of each clause. The category list was developed with input from Saudi legal experts to ensure alignment with the region's commonly used contract structures.

B. SYSTEM 2: LEGAL QA SYSTEM WITH RAG ARCHITECTURE

The second system is an interactive Question-Answering (QA) system designed to provide accurate, sourced answers to legal questions by leveraging a private knowledge base. It embodies a classic RAG architecture to overcome the limitations of a standalone LLM. The system is initialized once and then serves multiple queries through the following process:

1) A. INITIALIZATION & KNOWLEDGE BASE PREPARATION:

- 1) **Document Ingestion:** The system loads all .docx files from a specified directory (legal_documents). Each document is parsed, and its text content is extracted alongside metadata (e.g., the source file path).
- 2) **Chunking:** Documents are split into semantically coherent chunks using a `SemanticChunker`. This algorithm uses embeddings to break text at points of significant semantic shift, creating chunks that are more contextually meaningful than those from simple fixed-size splitting.
- 3) **Vector Store Creation:** Embeddings for each text chunk are generated using a pre-trained model (GATE-AraBert-v1) optimized for the Arabic language. These vector representations are stored and indexed in a persistent Chroma vector database. This database acts as the system's long-term memory.

2) B. QUERY PROCESSING & ANSWER GENERATION (RUNTIME):

For each user query, the following sequence is executed:

- 1) **Retrieval:** The user's query is converted into an embedding vector. This vector is used to query the Chroma database to find the most semantically similar text chunks from the knowledge base. The retriever is configured to return a small number of top candidates ($k=1$).
- 2) **Reranking (Advanced Retrieval):** An additional cross-encoder model (ARA-Reranker-V1) is employed to re-rank the retrieved documents. This model performs a more computationally intensive but accurate pairwise comparison between the query and each retrieved chunk, ensuring the absolute most relevant context is sent to the LLM.
- 3) **Context Truncation:** The retrieved context is truncated to fit within the LLM's context window token

limit, preventing overloading and ensuring efficient processing.

- 4) **Synthesis (Generation):** The retrieved context and the user's query are injected into a carefully designed prompt template. This prompt instructs the LLM (`Gemini 1.5 Flash`) to act as a legal assistant, answer concisely in Arabic, and base its answer **only** on the provided context. The model then generates an answer grounded in the retrieved documents.
- 5) **Output:** The system returns the generated answer along with the source metadata (e.g., the original document filenames) of the chunks used, providing crucial citations and verifiability.

All RAG-based legal consultations in this study were performed using Gemini 1.5 Flash as the generative model. Earlier drafts experimented with GPT-3.5-Turbo, but the final implementation and all reported results rely exclusively on Gemini 1.5 Flash. This clarification resolves any prior inconsistency.

3) EVALUATION SUBSYSTEM

The system includes a robust evaluation framework:

- It tests performance against a benchmark dataset (`all_dataset.json`) containing question/reference-answer pairs.
- It uses a cosine similarity metric computed over contextual embeddings generated by AraBERTv2, a BERT-base model pre-trained on large-scale Arabic corpora, to quantitatively compare the generated answer with the reference answer.
- It measures and records response times for performance benchmarking.
- It implements a caching mechanism to avoid reprocessing identical queries, saving computational resources.

The retrieval pipeline uses Gemini's native Arabic embedding model with a vector dimension of 768, which provides a balance between semantic accuracy and indexing efficiency. The final FAISS index contains approximately 18,400 vectors, corresponding to the chunked Saudi legal corpus. The total index storage size is 152 MB, including metadata and auxiliary structures. This configuration enables fast nearest-neighbour search while keeping memory overhead manageable for cloud deployment.

C. ALGORITHMIC ANALYSIS

The efficiency and scalability of an AI system are critical for its practical deployment. This section breaks down the two previously described systems into their core algorithmic components, provides pseudo-code for their key procedures, and calculates their computational complexity using Big O notation. This analysis provides insights into the performance characteristics and scaling behavior of each architecture.

1) ALGORITHM DERIVATION AND COMPLEXITY: SYSTEM 1 (NON-RAG)

- Overall Pipeline

The Non-RAG system functions as a sequential pipeline, primarily offloading its computationally heavy-lifting to the Gemini API. Its core algorithm can be summarized as follows.

Algorithm 1 Non-RAG Contract Analysis Pipeline

```

1: procedure AnalyzeContract(pdfPath, analysisType)
2:   text ← ExtractTextWithGemini(pdfPath)
3:   text ← CorrectDates(text)
4:   clauses ← ExtractClausesWithGemini(text)
5:   if analysisType is 'ambiguous' then
6:     result ← AnalyzeAmbiguityWithGemini(closures)
7:   else if analysisType is 'contradictory' then
8:     result ← AnalyzeContradictionWithGemini(closures)
9:   end if
10:  return result
11: end procedure

```

2) COMPLEXITY ANALYSIS OF SYSTEM 1

External API calls and the length of the input text dominate the complexity.

- **Time Complexity:** Let n be the number of characters/tokens in the input PDF.

- ExtractTextWithGemini, ExtractClausesWithGemini, Analyze*WithGemini: These are all calls to the external Gemini API. Their complexity is not directly governed by standard algorithmic rules but by the model's internal architecture (e.g., Transformer-based). The latency is approximately linear with the number of input tokens and output tokens, i.e., $O(n + m)$, where m is the number of output tokens. For complexity analysis relative to our system's input, we consider the dominant term: $O(n)$.
- CorrectDates: This involves a regular expression search over the text, which is $O(n)$.

Overall Time Complexity: $O(n)$. The pipeline is sequential, so the overall complexity is linear with the size of the document.

- **Space Complexity:**
 - The algorithm must store the extracted text $O(n)$, the list of clauses $O(n)$, and the final result JSON $O(m)$.
 - **Overall Space Complexity:** $O(n + m) \approx O(n)$, as n (input text) dominates m (output size).

3) ALGORITHM DERIVATION AND COMPLEXITY: SYSTEM 2 (RAG)

- Initialization Phase

The initialization phase is performed once to build the knowledge base.

Algorithm 2 RAG System Initialization

```

1: procedure InitializeSystem(forceReinit)
2:   embeddings ← LoadEmbeddingModel
3:   if forceReinit OR ¬VectorStoreExists then
4:     documents ← LoadDocuments/DocumentsDirectory)
5:     chunks ← SemanticChunking(documents, embeddings)
6:     vectorstore ← CreateVectorStore(chunks, embeddings)
7:   else
8:     vectorstore ← LoadVectorStore(embeddings)
9:   end if
10:  retriever ← SetupRetriever(vectorstore)
11:  llm ← SetupLLM
12:  qaChain ← CreateQChain(retriever, llm)
13:  return qaChain
14: end procedure

```

- Query Processing Phase

This phase is executed for every user query.

Algorithm 3 RAG Query Processing

```

1: procedure AnswerQuestion(qaChain, query)
2:   docs ← Retrieve(qaChain.retriever, query)
3:   context ← FormatContext(docs)
4:   truncatedContext ← TruncateContext(context)
5:   prompt ← FormatPrompt(truncatedContext, query)
6:   answer ← Generate(qaChain.llm, prompt)
7:   return {answer, GetSources(docs)}
8: end procedure

```

4) COMPLEXITY ANALYSIS OF SYSTEM 2

The complexity must be considered for both the one-time initialization and the per-query runtime.

- Initialization Complexity

Let D be the total number of documents, C be the average number of chunks per document, and E be the embedding dimension.

- **Time Complexity:**

- LoadDocuments: $O(D)$, linear in the number of documents.
- SemanticChunking: Requires generating embeddings for sentences/segments to find break-points. This is $O(D \cdot C \cdot E)$, often simplified to $O(D \cdot C)$ as E is constant.
- CreateVectorStore: Generating embeddings for all chunks and building the index (e.g., HNSW for Chroma) is $O(D \cdot C \cdot E)$. Index construction can be higher, but it is a one-time cost.
- **Overall Initialization Time Complexity:** $O(D \cdot C)$. This is a significant one-time cost.

- **Space Complexity:**

- The vector store must hold $D \cdot C$ vectors of dimension E .

- **Overall Initialization Space Complexity:** $O(D \cdot C \cdot E) \approx O(D \cdot C)$.

- Per-Query Runtime Complexity

Let $VC = D \cdot C$ be the total number of chunks in the vector store. Let K be the number of tokens in the retrieved context and L be the number of tokens in the generated answer.

• Time Complexity:

- **Retrieve:** Querying a vector database using an approximate nearest neighbor (ANN) search algorithm (e.g., HNSW) has complexity $O(\log(VC))$ on average.
- **Rerank:** The cross-encoder compares the query to each of the k retrieved documents. Its complexity is $O(k)$ per comparison, leading to $O(k)$ for k documents. Since k is small and fixed (e.g., $k = 10$), this is $O(1)$.
- **TruncateContext:** Tokenization and truncation are $O(K)$.
- **Generate:** The LLM generation step complexity depends on the model. For a Transformer decoder, generating L tokens with a context of length K is roughly $O(L \cdot K)$.
- **Overall Per-Query Time Complexity:** $O(\log(VC) + L \cdot K)$. This is efficient and scalable, as it depends logarithmically on the corpus size.

• Space Complexity:

- The memory is dominated by the loaded embedding model, LLM, and vector store index, which are constant after initialization.
- **Overall Per-Query Space Complexity:** $O(1)$ (amortized, excluding the pre-loaded models and index).

5) CONCLUSION

The complexity analysis reveals a fundamental trade-off between the two architectures. The **Non-RAG System** has no initialization cost but processes each document from scratch with linear time complexity $O(n)$. It is suitable for batch processing individual documents, but does not scale for querying a large corpus. The **RAG System** requires a significant one-time initialization cost $O(D \cdot C)$ to build its knowledge base. However, after this investment, the per-query time complexity is highly efficient, scaling logarithmically with the corpus size $O(\log(VC))$. This makes it ideal for interactive applications where an extensive knowledge base needs to be queried frequently. The intended use case should therefore guide the choice of system: frequent querying of a fixed corpus (favoring RAG) versus deep analysis of individual, unpredictable documents (favoring the Non-RAG approach).

VIII. EXPERIMENTS

To validate the proposed approach and assess its practical effectiveness, a series of experiments were conducted using real-world legal documents and advanced AI models. This section outlines the experimental design, data sources, and

evaluation metrics. We aim to demonstrate its reliability, accuracy, and potential for integration into digital legal consultation platforms. The experimental results provide valuable insights into the system's strengths, limitations, and opportunities for future development.

A. EXPERIMENTS DESCRIPTION

In this section, the different experiments carried out are elaborated. The baseline model experiments, along with the RAG-enhanced model experiments, are shown. Furthermore, contracts analysis model experiments are discussed.

1) HARDWARE SPECIFICATIONS

All runtime experiments were conducted using a cloud-based environment with the following specifications:

- CPU: Intel Xeon Platinum 8272CL (2.60 GHz)
- RAM: 32 GB
- GPU: Not utilized (Gemini API inference is cloud-hosted)
- Operating System: Ubuntu 20.04 LTS
- Network: Average API round-trip latency of 280–350 ms
- Gemini Backend: Google Vertex AI (Saudi-region datacenter where applicable)

As Gemini 1.5 Flash runs on Google's managed infrastructure, execution time includes both local preprocessing and network-based inference latency. All experiments were executed under identical network conditions to ensure fairness and reproducibility. Furthermore, To quantify cloud inference behavior, Gemini API latency was measured over 50 sequential requests under identical network conditions. The average round-trip latency was 312 ms, with a standard deviation of ± 41 ms, indicating stable performance across calls. These measurements reflect API-level latency only and do not include preprocessing or reranking time. Gemini 1.5 Flash supports an extended context window of up to 1 million tokens, but practical API constraints limit the size of individual requests. In our implementation, the effective usable window per request is approximately 32,000 tokens, including the prompt, retrieved legal text, and generated output. To stay well within this limit and ensure stable inference, we limit the total retrieved context to 10 chunks ($\approx 4,000$ – $6,000$ tokens). This design choice prevents truncation, ensures that all retrieved statutes remain intact, and motivates the need for the RAG pipeline with semantic chunking and reranking.

2) PROMPTS TEMPLATES SAMPLES

a) Legal Consultation (RAG) Prompt:

You are a Saudi legal assistant. Answer the user question strictly based on the retrieved legal text provided below.

DO NOT use external knowledge.

State the relevant Saudi regulation and article number in your answer. If the retrieved text is insufficient, state clearly.

Context:

retrieved_chunks

Question:

user_question

Provide your answer in clear and formal Arabic.

b) Contract Analysis Prompt:

You are a contract analysis expert. Analyze the following contract and return your response in valid JSON format.

For each clause, identify:

- 1) Clause text
- 2) Clause category (e.g., Payment, Termination, Obligations)
- 3) Risk level (Low, Medium, High)
- 4) Issues or missing elements
- 5) Whether it complies with Saudi regulations (True/False)

Contract text: contract_text

Return the result as a list of clause objects in JSON. Both templates were used consistently across all experimental runs.

3) BASELINE & RAG-ENHANCED MODEL EXPERIMENTS

This section details the experiment setting, software tools, and method we have used to assess AI models' performance in the legal advice and contracts analysis scenario. To serve as the driving force for the delivery of legal advice and analysis of legal documents, the Gemini 1.5 Flash model was utilized (as presented in Appendix B). We chose this model for its more powerful capabilities in context understanding and accurate results in legal text understanding. The system was implemented based on the Google Colab platform, which provides high-performance computing power as well as a GPU to run the model smoothly. Code was written and experiments were managed in a Visual Studio Code environment. The entire test process was built on Python and included a test framework and result analyzer. The evaluation was performed on a dataset of 150 legal questions, 50 being real-world scenarios from publicly available cases. This was done to mimic realistic legal advisory settings closely. The main goal was to evaluate the Gemini 1.5 Flash model to produce contextually correct legal answers and to compare it with the most advanced large language models, in particular Command-A, GPT-3.5, GPT-4, and ALLaM. All models were tested on the same question set under a similar experimental setup. All experiments used consistent LLM parameters to ensure reproducibility. Gemini 1.5 Flash was configured with:

- Temperature: 0.2
- Max Tokens: 1024
- Top-p: 0.95
- Top-k: 40
- Stop sequences: Arabic legal punctuation delimiters

A standardised prompt template was used for all legal questions:

“Provide a legally accurate answer in Arabic based strictly on the supplied context. Cite the article and regulation where applicable.”

For contract analysis, the prompt enforced JSON-structured output and mandatory clause-by-clause evaluation.

The RAG-augmented model investigated the effectiveness of the platform on providing an accurate and context-aware legal consultation using Gemini 1.5 Flash as the central language model. The performance of the model was compared with four state-of-the-art LLMs, including GPT-3.5, GPT-4, ALLaM, and Command. All evaluations were conducted on the test data, containing 100 questions and 10 major legal systems in Saudi Arabia, including 50 scenarios related to real legal problems. These were intended to evaluate the model's reasoning “If-Then-When” abilities, arithmetic performance, and simulation of real-world legal consultations. Users interacted with the platform by posting legal questions in free text. Implemented by Gemini 1.5 Flash, the Saudi Law Advisor was in charge of understanding all questions, searching for the most accurate legal content, and generating precise answers that were consistent with the Saudi legal body.

To ensure reproducibility and scientific rigor, all experiments were executed using a fixed set of parameters, standardized prompts, and consistent evaluation workflows. Baseline and RAG-enhanced results were validated through multiple runs, and all legal correctness judgments were verified by Saudi legal practitioners. These methodological steps were added to enhance transparency and strengthen the reliability of the reported findings.

4) CONTRACT ANALYSIS MODEL EXPERIMENTS

This paper tests the platform's ability to flag problematic clauses in legal contracts and suggest a course of action for compliance, primarily with the Gemini 1.5 Flash model, and with a few additional pre-trained language models, including GPT-3.5, GPT-4, ALLaM, and Command-A. The models were tested without additional training modifications on Ministry-specific data, except for the refinement process to assess the generalization of the models to analyze and align with Saudi legal standards.

We chose Gemini 1.5 Flash as the base model for this task, given its superior contextual awareness and legal passage comprehension. The platform was used to analyze different types of contracts. The tasks of the models were to check for problematic clauses, non-compliance, and recommendations for corrective action to comply with Saudi law.

To identify missing elements, each contract is evaluated against a predefined checklist of essential clauses typically required under Saudi commercial practice—such as payment terms, governing law, dispute resolution, termination conditions, liability allocation, confidentiality, and force-majeure provisions. Saudi legal practitioners validated this checklist. The system compares the extracted clauses with the checklist and flags any unmet items for non-compliance detection.

Each clause is evaluated using an LLM-assisted semantic analysis that checks for alignment with key Saudi legal requirements (e.g., compliance with the Saudi Commercial Law, Electronic Transactions Law, and Labour Regulations, where applicable). The system highlights clauses that contain prohibited terms, ambiguous obligations, or legally incomplete formulations and marks them as non-compliant. This hybrid rule-based and LLM-guided approach ensures robust and context-aware contract evaluation.

B. DATASET DESCRIPTION

Two datasets were used:

1) EVALUATION DATASET

The first dataset, used for testing purposes, contains 150 legal questions. These questions are distributed across ten primary legal domains, representing the major areas of Saudi law. Dataset Justification: The size of the evaluation dataset (150 questions and 50 real contract samples) reflects the limited availability of publicly accessible, high-quality Saudi Arabic legal cases. Saudi legal texts are not openly published in large-scale machine-readable form, and privacy regulations restrict broader release. Therefore, the dataset was curated with expert supervision to ensure domain validity rather than quantity. This approach aligns with prior work in Arabic legal NLP, where datasets are similarly small due to structural data scarcity. The size of the evaluation dataset (150 questions and 50 real contract samples) reflects the limited availability of publicly accessible, high-quality Saudi Arabic legal cases. Saudi legal texts are not openly published in large-scale machine-readable form, and privacy regulations restrict broader release. Therefore, the dataset was curated with expert supervision to ensure domain validity rather than quantity. This approach aligns with prior work in Arabic legal NLP, where datasets are similarly small due to structural data scarcity.

The legal fields covered are tabulated in Table 2.

This diversity ensures that the AI system is rigorously tested on a broad range of legal topics relevant to Saudi law.

2) TRAINING/RETRIEVAL CORPUS

For the platform’s operational training, the second dataset is based on the legal corpus provided by the Saudi Government’s Expert Authority [29]. This dataset consists of 640 legal scenarios, which represent real-life legal cases across various branches of Saudi law. These scenarios, presented as narratives of individuals involved in legal matters, cover a wide array of issues.

C. EVALUATION METRICS

These models were benchmarked to evaluate their performance and capabilities in legal text processing. Performance was assessed using the four metrics. The **Accuracy** refers to the percentage of legally correct responses. **Response Time** calculates the time taken to generate an answer.

TABLE 2. Legal domains used in experiments.

Legal Domain	Key focus area
Basic and Constitutional Laws	Foundational legal principles and governance of Saudi Arabia
Commercial and Investment laws	Business transactions and investment-related regulation
Financial and Taxation Laws	Taxation frameworks and financial policies.
Labor and Employment Laws	Employee rights, employer obligations, and workplace regulations.
Real Estate and Construction Laws	Property ownership, real estate transactions, and construction-related matters.
Criminal and Penal Laws	Criminal offenses and corresponding penalties.
Family and Personal Status Laws	Marriage, divorce, inheritance, and child custody.
Health and Medical Laws	Healthcare regulations, patient rights, and medical malpractice.
Environmental, Water, and Energy Laws	Environmental protection and energy regulations.
Telecommunications and Technology Laws	Legal aspects of digital technologies and telecommunications.

Textual Similarity is a score between the generated and a reference answer (>70% was considered correct) and expert evaluation that includes manually reviewing and scoring of results by Saudi lawyers to ensure accuracy and contextual appropriateness.

D. EXPERT EVALUATION CRITERIA

To ensure a legally meaningful assessment of the generated answers, we used a structured review process involving three Saudi legal experts, each with at least five years of legal practice. The experts evaluated model outputs using the following rubric:

- 1) Legal Correctness (0–5 points): Accuracy of the answer with respect to Saudi statutory and regulatory law.
- 2) Clarity and Structure (0–5 points): Whether the answer is easy for non-lawyers to understand, logically structured, and free from ambiguity.
- 3) Regulation Alignment (0–5 points): Degree to which the answer explicitly cites or aligns with relevant legal articles and provisions.
- 4) Practical Utility (0–5 points): Whether the response provides actionable guidance consistent with typical legal advisory practice.

Each answer could therefore receive a maximum of 20 points per expert. Scores were averaged across reviewers. When expert scores differed by more than 2 points, a consensus discussion was held to standardize the final rating. This structured rubric ensures transparent and reproducible expert scoring.

E. BIAS AND FAIRNESS CONSIDERATIONS

To minimize potential bias in model responses, several safeguards were incorporated during dataset preparation and evaluation. First, the Saudi legal corpus and question dataset were reviewed to ensure neutrality with respect to

gender, nationality, and socioeconomic background, avoiding examples that could disproportionately favor or disadvantage specific demographic groups. As traditional fairness metrics such as demographic parity and equalized odds are not directly applicable to legal consultation tasks—where responses depend solely on statutory text rather than user attributes—we adopted alternative domain-appropriate checks. These included consistency testing across hypothetical user profiles, ensuring that equivalent legal questions generate identical outputs regardless of the assumed identity of the inquirer. Additionally, all generated answers were reviewed by Saudi legal experts, who verified that the system’s reasoning remained grounded in law rather than user-specific assumptions. These measures collectively enhance fairness and reduce the likelihood of biased or discriminatory legal recommendations.

F. LEGAL RELIABILITY AND VERIFICATION MECHANISM

To ensure that all generated legal responses are trustworthy and verifiable, the system implements multiple reliability safeguards. First, every answer is strictly grounded in retrieved statutory or regulatory text through the RAG pipeline, preventing hallucinations and ensuring factual alignment with Saudi law. Second, the prompt template enforces citation behavior, requiring the model to reference the article and regulation used in its reasoning. Third, a post-generation validation layer checks whether the cited articles actually appear in the retrieved context; responses failing this check are automatically rejected and regenerated. Additionally, the system restricts generative freedom by instructing the model to avoid opinion-based or speculative statements. Finally, the output specifications and legal interpretations were reviewed by Saudi legal experts, ensuring that the system consistently produces legally reliable, verifiable, and context-appropriate guidance.

IX. RESULTS

The evaluation shows a comparison of six legal interpretation models on three main aspects: accuracy, average response time, and expert ratings. Gemini 1.5 Flash provided the best overall fit for all the models tested. It attained a maximum accuracy of 98.67% and answered 148 of 150 questions correctly. It also achieved an excellent average response time of 1.63 seconds and a quality expert score of 149 points. Close behind was GPT-3.5, where the accuracy is 98.00% (147 correct answers). It was even faster than it, with 1.52 seconds per query on average, but it scored 123 points lower as measured by the experts. GPT-4 did well also, it was 97.33% accurate with 146 correct responses, had an average response time of 2.56 seconds, and was rated by the experts at a rating of 146 points. ALLaM came in second with a precision of 96.00%, counting 144 correctly answered questions, and a response average of 4.51 seconds. It scored 124 points from expert evaluation. Task A, despite having a little lower accuracy, did not have the lowest performance speed, except for the L4, but still it was used for the comparison among the

commercial systems, with the Highest correct answer rate 141 (94.00%) and the lowest responding time 6.38 seconds. And interestingly, even though it was slower, it had a relatively high 145-point expert rating. The results are summarized in the Tables 3 and 4 and a comparison between Baseline and Rag systems is presented in the Table 5.

A. RESULTS OF LEGAL CONSULTATION PERFORMANCE USING THE RAG MODE

Aside from the baselines, all models were tested with the Retrieval-Augmented Generation (RAG) approach to increase the precision and quality of answers, integrating legal texts. The incorporation of RAG resulted in a tangible quality and completeness boost of the generated outputs. The Gemini 1.5 Flash gained the best results in general. It continuously generated full, well-structured, and legally accurate responses with an average similarity of 96.57%. It was also sensitive and specific, and provided the responses with an average response time of 143.52 seconds. The RAG processing time reported in the experiments (143.52 seconds for Gemini) reflects the full evaluation pipeline, not the operational latency of the deployed legal advisor. During experimental testing, each query triggered complete semantic retrieval, embedding generation, cross-encoder reranking, context compression, and detailed logging. These steps were executed sequentially for every test question without caching to ensure reproducibility and traceability. In the deployed system, retrieval and reranking operations are cached, embeddings are pre-computed, and only the final generative step is executed—resulting in typical user-facing response times that fall within normal interactive ranges. The reported latency therefore represents internal evaluation cost rather than real-time system performance. To mitigate random fluctuations, each response-time measurement was averaged over three independent runs per model. Variance across runs remained low ($\pm 4.3\%$ on average), indicating stable timing behavior. Although formal significance testing is limited by dataset size, repeating the experiments strengthens the reliability of the reported averages. In contrast to the above, Command A has a little better similarity score (97.11%), but the running time was much greater (280.67 seconds). GPT-4 and GPT-3.5 resulted in much longer response times (264.06 seconds and 211.98 seconds, respectively), but were less accurate as their similarity scores are 88.31%, 93.12%, respectively.

While some of the models gave reasonably good answers, Gemini 1.5 Flash proved to be the most reliable and practical choice for high-quality advice in law (enhanced with RAG), thanks to higher accuracy, depth, and structural consistency. Due to the limited availability of large-scale Arabic legal datasets, full statistical significance testing was not feasible. However, to improve the reliability of the reported results, all RAG and non-RAG experiments were repeated three times, and average values were recorded. Variability across runs remained consistently low, typically within a 4–5% margin. Although formal hypothesis testing could not be

TABLE 3. Performance comparison of legal consultation models (Baseline).

Model	Accuracy (%)	Correct Answers (/150)	Response Time (s)	Expert Rating (/150)
Gemini 1.5 Flash	98.67	148	1.63	149
GPT-3.5	98.00	147	1.52	123
GPT-4	97.33	146	2.56	146
ALLaM	96.00	144	4.51	124
Command-A	94.00	141	6.38	145

TABLE 4. Contract analysis performance across models.

Model	Total Clauses	Correct	Incorrect	Quality Assessment
Gemini 1.5 Flash	7	6	1	Coherent, accurate issue statements aligned with legal principles
ALLaM	9	7	2	Repetitive content, lacks structural organization
Command-A	5	4	1	Limited depth and scope compared to Gemini
GPT-4	6	5	1	Merges multiple legal issues, obscuring analysis

*** 3 رفم ***

**** التبدد الضامض:** حدد أيام العمل العادية بـ 5 أيام في الأسبوع ويحدد ساعات العمل بـ 8 يومياً ويلتزم الطرف الأول بأن يدفع للطرف الثاني أجراً اضافياً عن ساعات العمل الإضافية بوزن أجر الساعة مضاعفاً إليه 50% من أجره الأساسي.

**** المشكلة في البدن:** يقول "8 يومياً" وهنا عبر منطقي، يُتراض أن المفقود هو 8 ساعات يومياً، يجب توضيح طريقة حساب الأجر الإضافي (هل هو لكل ساعة إضافية فوق 8 ساعات أم لكل يوم عمل إضافي؟ وما هو تعريف "ساعة العمل الإضافية")؟

**** التوصلات:** ** تُحدد أيام العمل العادية بـ 5 أيام في الأسبوع، وعدد ساعات العمل بـ 8 ساعات يومياً. يُتفق للمُعرف الثاني أجر إضافي عن كل ساعة عمل إضافية تتجاوز 8 ساعات يومياً، بواقع 150% من أجره الأساسي للساعة الواحدة.

*** 5 رُفُ ***

***البند الخامس:** يبحق الطرف الثاني عن كل عام اجارة مدفوعة منها 22 يوما مدفوعة الاجر ويحدد الطرف الأول تاريخها خلال سنة الاسحقاق وفقاً لطرف العمل على ان يتم دفع ايجار الاجارة مقدما عند استحقاقها وللطرف الأول تأجيل الاجارة بعد نهاية مدة استحقاقها لمدة لا تزيد عن 90 يوماً كما له بموافقة الطرف الثاني كتابة تأجيلها إلى نهاية السنة التالية لسنة الاسحقاق وذلك حسب مقتضيات ظروف العمل.

*** المشكلة في البلد ** البلد عامض في تحديد كيفية تحديد تاريخ الإجازة، وخاصةً آلية التأجيل. يجب توضيح المزيد من التفاصيل حول حق الطرف الأول في التأجيل.

***التوصيات: يستحق الطرف الثاني إجازة مدفوعة الأجر لمدة 22 يوم عمل عن كل سنة كاملة من العمل . يحق للطرف الناجم تقديم طلب تحديد موعد إجازته للطرف الأول قبل 30 يوماً على الأقل . يحق للطرف الأول تأجيل

تحليل الحف و تحديد البود الخامضه:

1 فح

*** البلد الغامض: ** هذا العدد 1 سنة يبدأ من تاريخ 2009-03-24، وينتهي في 2015-03-19 علماً بأن تاريخ منافسة الطرف الثاني للحل هو 2022-03-20. وتحدد لمدة أو لمدة مماثلة ما لم ينص أحد الطرفين الآخر خطياً بعدم رعيه في التجديد قبل (60) يوماً من تاريخ انتهاء العدد.

***المشكلة في البند **بأنضمن البند نفاصاً واصحاً في تواريخ البدء والانتهاء تاريخ البدء (2024-03-2009) عبر مطلق، وينبغي خطأ مطبعياً. كما أن تاريخ الانتهاء (2015-03-19) سابق لتاريخ بدء العمل (20-03-2022).
يعتبر البند إلى الوضوح بشأن آلية التجديد، فهل التجديد لمدة سنة واحدة فقط، أم يمكن أن يكون لمدة مختلفة؟

*****التوصيات***** : بعد هذا العقد سنة واحدة تبدأ من تاريخ 2022-03-20 وتنتهي في 2023-03-19، ينجذ السيد بولمنا على الأهل من تاريخ انتهاء السنة الحالية للعقد.

*** رَفْع 2 ***

*** البند الخامس: ** يخضع الطرف الثاني لفترة تجريبية مدتها 90 يوماً تبدأ من تاريخ مبادرته للعمل ولا يدخل في حسابها إجازة عديي الفطر والأضحى والإجازة المرضية ويكون الطرفان الحق في إنهاء العقد خلال هذه الفترة، مالم ينص العقد على أحدهما في الإنهاء

*** المشكلة في البند **: الفقرة الأخيرة "مالم ينص العقد على أحقية أحدهما في الإنهاء" عاصية وغير واضحة، ما المقصود بـ "مالم ينص العقد"؟ هل هذا يعني استثناء ما؟ يجب توضيح شروط إنهاء العقد خلال فترة التجربة بشكل صريح.

*** النوصيات: ** "يضع الطرف الثاني لغيره تجربة منها 90 يوماً موائلاً لنا من تاريخ مباشر للعمل لا تحسب إجازات عبد العطر والأضحي والإجازات المرضية ضمن هذه المدة. يحق لكل من الطرفين إنهاء العقد خلال فترة التجربة دون سبب ودون أي نوصي".

FIGURE 5. Results of Legal Contract Analysis Model Sample 1.

applied, these repeated trials strengthen the reproducibility and stability of the reported findings.

B. RESULTS OF LEGAL CONTRACT ANALYSIS MODEL

When comparing performance across the different models for legal contract analysis, the findings showed significant differences in the capability of distinct models to recognize defective clauses. After examining the output of Gemini 1.5 Flash, we noticed that the model returned seven clauses, with six clauses together forming a coherent, accurate issue statement aligned with legal principles and the erroneous first clause. This is indicative of its accuracy and reliability.

as we can see in the detailed clause-by-clause evaluation on Gemini’s output shown in Figures 5 and 6.

In contrast, ALLaM generated an analysis consisting of nine clauses, where the first clause was incorrect, the second and third were accurate, and the remaining provisions were also correct. However, the presence of repetitive content diminished the overall quality of the analysis and indicated a lack of structural organization.

Command A generated five clauses, with the first one being an incorrect clause, and the remaining four were correct. Although correct, such clauses did not have the sort of depth and scope as evident in Gemini.

الإجازة لمدة لا تتجاوز 90 يوماً بعد انتهاء السنة الاستحقاقية، بشرط موافقة الطرف الثاني كتابياً. يحق للطرف الأول أيضاً، بموافقة كتابية من الطرف الثاني، تأجيل الإجازة إلى نهاية السنة التالية، وفقاً لمسؤوليات العمل".

بند رقم 8:

البند الخامس: "تسحق الموظفة إجازة وضع بأجر كامل لمدة عشرة أسابيع توضعها كيف تشاء تبدأ بعد أقصى بأربعة أسابيع قبل التاريخ المرجح للوضع ويحدد التاريخ المرجح للوضع بواسطة طبيب المنشأة أو بناء على شهادة طبية صادقة من جهة صحية

** المشكلة في البند **:"التعبير "توضعها كيف تشاء" قد يكون عامضاً ويحتاج إلى توضيح حول كيفية طلب الإجازة وتقسيمها.

** التوضيحات:** "تسحق الموظفة إجازة وضع مدفوعة الأجر لمدة 10 أسابيع. يجب على الموظفة إبلاغ الطرف الأول بتاريخ وضعها المتوقع، مدعوماً بشهادة طبية من جهة صحية معتمدة، قبل أربعة أسابيع على الأقل من التاريخ المتوقع للوضع. تُوزع إجازة الوضع على فترات متصلة أو منفصلة حسب اتفاق الطرفين، مع مراعاة عدم تجاوز المدة الإجمالية البالغة عشرة أسابيع".

بند رقم 28:

البند الخامس: "مدة هذا العقد لمدة سنة واحدة ميلادية واحدة تبدأ من تاريخ مباشرة الموظف للعمل لدى الشركة وتنتهي بعد العمل لمدة سنة ميلادية من تاريخ المباشرة الفعلية للعمل وتجدد لمدة أو لمدد مماثلة ما لم ينسحب أحد الطرفين الآخر خطياً بمرور سنة في التاريخ قبل سبعة أيام من تاريخ انتهاء العقد وفقاً لأحكام المادة 123 من هذا العقد

** المشكلة في البند **:"الفقرة "وتنتهي بعد العمل لمدة سنة ميلادية من تاريخ المباشرة الفعلية للعمل" عامضة وغير دقيقة، ما المقصود بـ "بعد العمل لمدة سنة ميلادية"؟ يجب أن تكون الصياغة أوضح بشأن تاريخ الانتهاء. كما أن الإشارة إلى "أحكام المادة 123" تدور حول تحديد مصدر هذه المادة تجعل البند عامضاً.

** التوضيحات:** "مدة هذا العقد سنة ميلادية واحدة تبدأ من تاريخ مباشرة الموظف للعمل وتنتهي بعد مرور سنة ميلادية كاملة من تاريخ مباشرة العمل. يتجدد العقد تلقائياً بنفس الشروط لمدة سنة ميلادية أخرى ما لم يخطر أحد الطرفين الآخر خطياً بمرور سنة في عم الجديد قبل سبعة أيام من تاريخ انتهاء السنة الحالية للعقد".

بند رقم 33:

البند الخامس: "يلزم الطرف الثاني بالعمل من خلال وريثات بأوقات متفرقة يتم إتمامها من قبل الطرف الأول

** المشكلة في البند **:"يقضي البند إلى تحديد آلية اعتماد الوريثات المتفرقة، ما هي الشروط والإجراءات؟ هل هناك حد أدنى لساعات الراحة؟

** التوضيحات:** "يلتزم الطرف الثاني بالعمل وفقاً لجدول وريثات متفرقة يحددها الطرف الأول ويُبلغ بها الطرف الثاني قبل [عدد] يوم/أيام من بداية كل دورة عمل. يجب أن يتضمن جدول الوريثات الحد الأدنى لـ [عدد] ساعة راحة بين كل وريثة وأخرى، وفقاً لقانون العمل".

البند المبتدعة(6): 7، 9، 11، 17، 19، 20، 21، 22، 27، 29، 32: هذه البنود واضحة بشكل عام، لكن بعضها قد يُحسن من خلال إضافة تفاصيل أكثر. على سبيل المثال، يمكن تحديد مصادر المعلومات القانونية بدقة (مثل رقم المادة من نظام العمل) في البنود التي تشير إلى مواد قانونية. يُنصح بمراجعة هذه البنود وتحديثها لتكون أكثر دقة وشمولاً.

FIGURE 6. Results of Legal Contract Analysis Model Sample 2.

In the case of GPT-4, there were six clauses, the first of which was wrong. Furthermore, the third factor joined together three unique legal issues into one, which obscured and narrowed the analysis.

With such an examination, the Gemini 1.5 Flash emerges as the most robust and effective model for legal contract analysis, surpassing all other models in the legal contract analysis in terms of accuracy and clause consistency, while forgoing redundancies or grand merging—results reinforced by the data in Figures 5.1 and 5.2.

TABLE 5. Runtime comparison of legal consultation models: Baseline vs. RAG.

Model	Baseline Response Time (s)	RAG Response Time (s)
Gemini 1.5 Flash	1.63	143.52
GPT-3.5	1.52	211.98
GPT-4	2.56	264.06
ALLaM	4.51	—
Command-A	6.38	280.67

C. ERROR ANALYSIS AND FAILURE CASES

Despite the overall strong performance of the proposed system, several failure cases were observed during evaluation. The most common error type arises from incomplete retrieval when relevant legal clauses span multiple chunk boundaries, leading to partial or missing context and occasionally incomplete answers. A second category of errors involves clause misclassification during contract analysis, particularly for complex or multi-purpose clauses that span payment,

liability, and confidentiality elements simultaneously. In such cases, the model sometimes assigns the clause to only one category rather than multiple applicable categories. Third, in a limited number of responses, the model provided a statutory citation that did not appear in the retrieved context, typically because the question was ambiguous. These failure cases highlight areas where improved chunking strategies, multi-label clause classification, and enhanced citation validation could further strengthen system reliability.

X. CONCLUSION AND FUTURE WORK

The findings unveiled the immense potential of AI, including LLMs (eg, Google's Gemini 1.5 Flash) for the development of legal service delivery in Saudi Arabia's legal environment. Having combined the Gemini system with the RAG (Retrieval-Augmented Generation) technique and employed a dedicated Saudi legal corpus, the developed system also quite successfully tackled some of the practical issues in legal counseling and contracts analysis. The system exhibited high accuracy and efficiency levels, as verified by empirical experiments and expert legal evaluations, which affirm that the system is in line with the Saudi laws. And the service has demonstrated its real-world usefulness in examining complex Arabic legal documents and accurately spotting contractual risks, providing accessible and dependable tools for individuals and businesses that remain unattended by traditional legal services. Comparison results across baselines, including channel-bond GPT-4, GPT-3. 5, Control-A, and ALLaM—

also contributed to the conclusion that the performance of Gemini outperformed other specialized systems.

Nevertheless, certain limitations were identified. The model's performance when dealing with extremely difficult or ambiguous legal texts that are not well represented in the dataset can be different. Even if RAG is adopted, the limitation to limited legal databases may limit the adaptability of the method to the new and fast-changing regulations. Moreover, the LLM models themselves contain biases and may also introduce errors, hence the ongoing need for monitoring and fixing to ensure reliability and fairness.

As far as the limitations of the proposed system, we notice that the Legal data is generally unstructured and needs to be formatted to fit the AI model. The data has to be pre-processed and cleaned up before it can be used for training. Besides, legal gaps may differ depending on the contract type and language, and the system must therefore be trained to identify multiple types of gaps. Furthermore, the solution shall have to be very accurate in picking up on backdoors without producing false alarms, so it must be tested and verified over and over again.

Despite the safeguards incorporated into the proposed system, several inherent limitations of LLM-based legal interpretation remain. First, legal language often contains ambiguous or multi-layered meanings that require contextual, situational, or judicial interpretation, which a text-only model cannot fully capture. Second, LLMs are susceptible to hallucinations when the retrieved statutory context is incomplete or when questions contain implicit assumptions not supported by law. Third, generative models are non-deterministic, meaning that identical inputs may produce slightly different outputs across runs. Fourth, the system relies heavily on the quality and coverage of the RAG corpus—if a relevant regulation is missing, the model cannot compensate by relying on external knowledge. Finally, the model cannot replace professional legal judgment and should be treated as an assistive tool rather than an authoritative legal decision-maker. It is essential to clarify that the system provides legal information, not legally binding advice. All responses are generated by interpreting statutory text retrieved through the RAG pipeline and do not account for case-specific facts, procedural nuances, or individualised legal strategies. The system is not a substitute for a licensed attorney, and users are explicitly instructed to consult qualified legal professionals for any decisions that may carry legal or financial consequences. This design ensures compliance with advisory limitations while still offering accurate, statute-grounded informational guidance.

Indeed, the restrictions just discussed are an avenue of investigation to automate the process. Moreover, the following steps comprise improving the platform for more complex legal use cases, deeper analysis of the contracts, and the reduction of errors and biases. Planned follow-up work involves the complete automation of contract generation according to the legal framework (e.g., law regulations, company internal policies), as well as the extension to

additional types of documents, such as internal policies and regulations, to address corporate and enterprise requirements properly. Additionally, user experience will be enhanced as better and easier-to-use interfaces are developed to protect the confidentiality and security of users' legal and private data.

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